

Tutorial No : 3

Tutorial Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

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Program:

```
#include <iostream>

void readArray(int arr[], int n) {
    for (int i = 0; i < n; ++i) std::cin >> arr[i];
}

void displayArray(const int arr[], int n) {
    for (int i = 0; i < n; ++i) std::cout << arr[i] << " ";
    std::cout << "\n";
}

void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

void sortArray(int arr[], int n) {
    for (int i = 0; i < n - 1; ++i)
        for (int j = 0; j < n - i - 1; ++j)
            if (arr[j] > arr[j + 1])
                swap(arr[j], arr[j + 1]);
}
```

```
}  
  
int main() {  
    int n;  
    std::cout << "Enter size: ";  
    std::cin >> n;  
    int arr[n];  
    std::cout << "Enter " << n << " elements:\n";  
    readArray(arr, n);  
    sortArray(arr, n);  
    std::cout << "Sorted: ";  
    displayArray(arr, n);  
    return 0;  
}
```

Output:

Output Clear

```
Enter size: 4  
Enter 4 elements:  
12  
45  
12  
5  
Sorted: 5 12 12 45  
  
--- Code Execution Successful ---
```